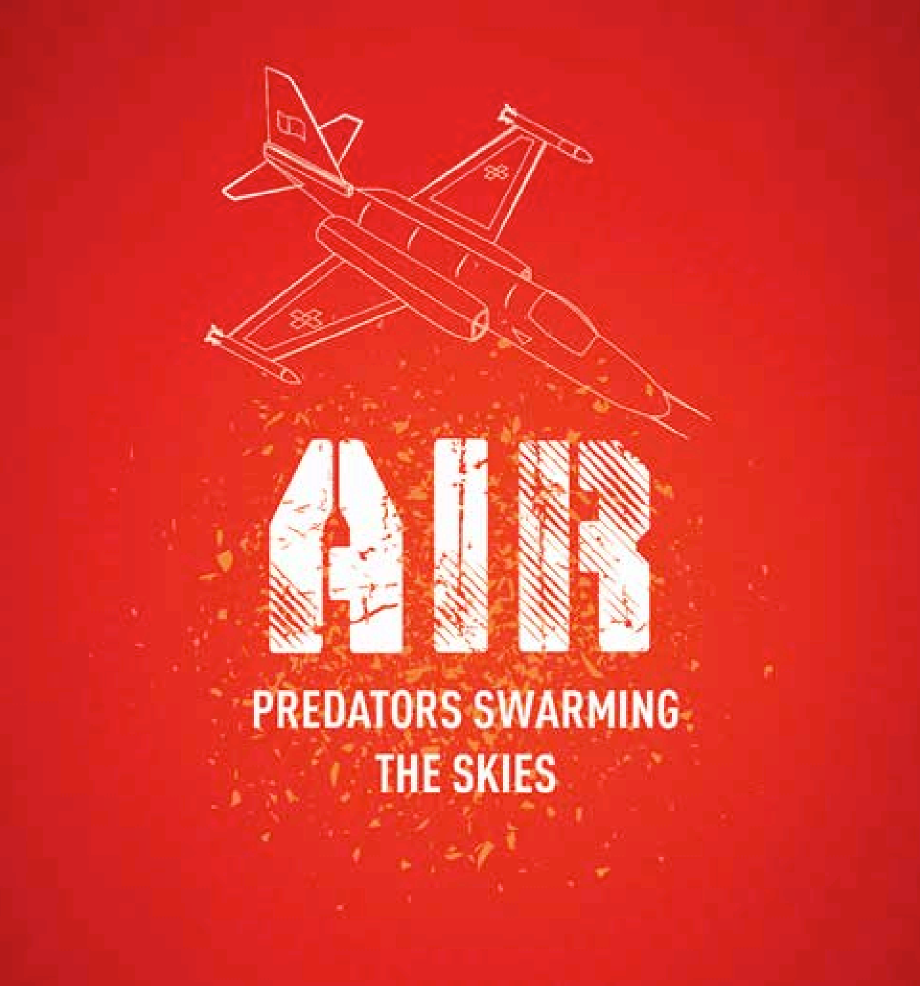

Technically known as unmanned combat aerial vehicle (UCAV), combat drones have a long and established history of use by the military. One of the earliest concepts of a combat drone appeared in the December 1940 issue of Popular Mechanics, which presented plans for a drone with a propellor loaded with "mechanical eye" which is a camera, a television transmitter, an explosive payload and a radio receiver for controlling the wings and firing the payload. This device called the "robot television bomber" could be remotely operated by personnel on the ground, or even from a plane that followed the drone or flew high above it. In 1948, the US Air Force commissioned drones for target practice from the Ryan Aeronautical Company, which resulted in the development of the Ryan Firebee target drones by 1953. These drones were so good at their jobs, that through a series of improvements, they continue to be used for target practice till this day.

In 1973, nuclear physicist and model airplane hobbyist John Stuart Foster Jr built two prototypes for the DARPA, powered by lawn mower engines that would go on to become the precursors to modern combat drones. In the same year, in 1973, Israel used the unmanned and unarmed Ryan Firebee drones acquired from the US to provoke Egypt into emptying up its entire arsenal of anti aircraft missiles. However the efficacy of the Soviet built surface to air "SAM" missiles was devastating to the Israeli air force, spurring the US into initiating its stealth aircraft program to defeat these SAM missiles. Israel went on to develop their own glider-like drones in the 70s and 80s, pioneering the use of UAVs for surveillance, decoys and warfare, which is the approach used today by most combat drones. These drones managed to neutralise the entire Syrian air defence SAM network, at the outset of the 1982 Lebanon War, in an incident known as the "Bekaa Valley Turkey Shoot".

The US began developing its own drones based on the Israeli models, and their successful use in the Gulf War spurred militaries around the world to develop and adopt the technology. The US used a number of drone strikes during the ongoing war on terror, which has incidents where civilians became the targets of attacks, that goes to show that even with a team of remote control operatives, the killer robots have collateral damage. Development of the combat drones are being infused with intelligence and autonomy, to improve the reactiveness and decision making process. These are supported by specialised vehicles to carry and deploy the drones, including seafaring drone aircraft carriers, amphibious vehicles, and dropships that can release swarms of weaponized drones at once.

In March 2020, a Kargu 2 drone made by the Turkish company STM, loaded with an explosive payload was deployed in Libya by the government forces to tackle a breakaway faction. Once deployed, it hunted down a human target and attacked in a kami-kaze style rush. It is unclear if there were any casualties, but the incident may have been the first time that a lethal autonomous weapons system attacked a human. The available details were outlined in a UN report in May 2021, with information on the attack and its implications published in the New Scientist magazine. When it comes to enforcing an international treaty on killer robots, this incident clearly demonstrates that it may already be too late.

Credit: STM

The Kargu 2 drone, may just be the first killer robot.

Quadcopters are available for cheap for a number of civil uses, and these have